Final report

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1 Introduction

1.1 Research Question

Does measles vaccination coverage among students in kindergarten through grade 12 differ between public and private schools in New York State (excluding New York City) during the 2018–2019 school year?

1.2 Scientific Plausibility and Context

Measles is one of the most contagious human diseases and can lead to serious illness and death. Vaccination is highly effective, with two doses of the measles vaccine providing approximately 97% lifelong protection (Disease Control and Prevention 2025). However, because measles spreads so easily and can cross borders rapidly, outbreaks can occur in any community where vaccination coverage falls below 95% (Organization 2024).

Variation in vaccination coverage across schools is a critical public health concern. Previous research has shown that immunization exemption rates are significantly higher in private schools than in public schools in the United States, suggesting that children attending private schools may be at greater risk of vaccine-preventable diseases (Shaw et al. 2014). Higher exemption rates can lead to lower vaccination coverage and may create pockets of susceptibility that increase the risk of measles outbreaks. Therefore, identifying differences in vaccination coverage across school types can help inform targeted interventions and outreach strategies aimed at maintaining herd immunity and preventing vaccine-preventable disease outbreaks.

2 Materials and Methods

2.1 Data

The data were obtained from the New York State Health Data portal and were provided by the New York State Department of Health. The School Immunization Survey contains school-level aggregate information on student immunization status for grades kindergarten through 12 in New York State, excluding New York City public schools.

To match the research question, the data were restricted to the 2018–2019 reporting period and to schools classified as either public or private. Measles vaccination coverage was the main outcome variable. Because this variable was recorded as a percentage string in the raw dataset, the percent sign was removed and the values were converted to numeric format before analysis. Schools with missing measles vaccination coverage were excluded from the final analytic sample.

```r
# Load file
School_immun_data <- read.csv("../data/school_immun_2012_2019.csv")

# Function to convert percentage strings to numeric values
pct_to_num <- function(x) {
  x <- gsub("%", "", x)
  x[x == ""] <- NA
  as.numeric(x)
}

# Subset data to the 2018-2019 school year
immun_1819 <- School_immun_data[School_immun_data$Report.Period == "2018-2019", ]

# Keep only Public and Private schools
immun_1819 <- immun_1819[immun_1819$Type %in% c("Public School", "Private School"), ]

# Convert measles vaccination coverage from percentage to numeric
immun_1819$measles_pct <- pct_to_num(immun_1819$Percent.Immunized.Measles)

# Remove observations with missing vaccination coverage
immun_1819 <- immun_1819[!is.na(immun_1819$measles_pct), ]

# Create datasets by school type
public_schools  <- immun_1819[immun_1819$Type == "Public School", ]
private_schools <- immun_1819[immun_1819$Type == "Private School", ]
```

2.2 Statistical Test

2.2.1 Assumptions for Welch Two-Sample t-Test

1. Independence of observations

Each row represents one separate school, and vaccination coverage in one school is not a duplicate measurement of another school. This assumption is satisfied.

2. Continuous outcome variable

Measles vaccination coverage was converted from a percentage string to numeric format. This assumption is satisfied.

3. Normality

```
# Shapiro-Wilk tests for normality
shapiro.test(public_schools$measles_pct[1:500])
shapiro.test(private_schools$measles_pct[1:500])
```

The Shapiro–Wilk test showed significant departures from normality for both public schools (W = 0.529, p < 2.2e-16) and private schools (W = 0.538, p < 2.2e-16), indicating that measles vaccination coverage was not normally distributed in either group.

The normality assumption was therefore not met. However, Welch's two-sample t-test was retained as the primary analysis because both groups had large sample sizes, allowing the Central Limit Theorem to support the validity of the test. A non-parametric Wilcoxon rank-sum test was additionally conducted as a sensitivity analysis.

2.2.2 Primary Analysis: Welch Two-Sample t-Test

```
welch_result <- t.test(measles_pct ~ Type,
                       data = immun_1819,
                       var.equal = FALSE)
welch_result
```

The Welch two-sample t-test indicated a statistically significant difference in mean measles vaccination coverage between public schools (mean = 98.85%) and private schools (mean = 88.93%), with t = -18.04, df = 1954.2, p < 2.2e-16. The 95% confidence interval for the mean difference ranged from -11.00 to - 8.84 percentage points.

2.2.3 Sensitivity Analysis: Wilcoxon Rank-Sum Test

```
wilcox_result <- wilcox.test(measles_pct ~ Type,
                             data = immun_1819,
                             exact = FALSE)
wilcox_result
```

The Wilcoxon rank-sum test also showed a statistically significant difference between the two groups (W = 2,146,194, p < 2.2e-16), consistent with the primary Welch t-test result.

2.3 Data Availability

All code, data, and references needed to reproduce this report are available at: https://github.com/pinyuezou-glitch/Cornell_VTPEH6270.git

3 Results

3.1 Descriptive Statistics

```r
# Summary statistics by school type
desc_stats <- data.frame(
  School_Type = c("Public School", "Private School"),
  N = c(nrow(public_schools), nrow(private_schools)),
  Mean = c(round(mean(public_schools$measles_pct), 2),
           round(mean(private_schools$measles_pct), 2)),
  Median = c(round(median(public_schools$measles_pct), 2),
             round(median(private_schools$measles_pct), 2)),
  SD = c(round(sd(public_schools$measles_pct), 2),
         round(sd(private_schools$measles_pct), 2)),
  Min = c(round(min(public_schools$measles_pct), 2),
          round(min(private_schools$measles_pct), 2)),
  Max = c(round(max(public_schools$measles_pct), 2),
          round(max(private_schools$measles_pct), 2))
)

knitr::kable(desc_stats,
             col.names = c("School Type", "N", "Mean (%)", "Median (%)",
                           "SD", "Min (%)", "Max (%)"),
             caption = "Descriptive Statistics of Measles Vaccination
             Coverage by School Type (2018-2019)"
             )
```

Table 1: Descriptive Statistics of Measles Vaccination Coverage by School Type (2018–2019)

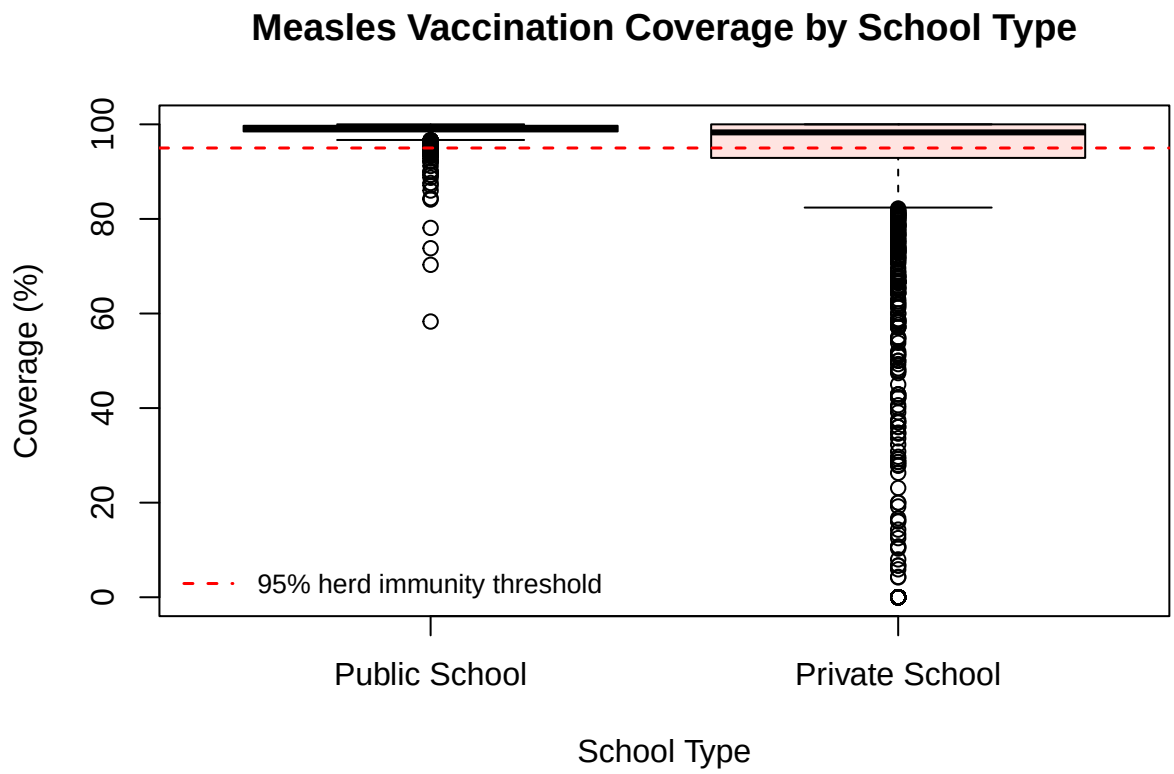
School Type	N	Mean (%)	Median (%)	SD	Min (%)	Max (%)
Public School	2834	98.85	99.2	1.7	58.3	100
Private School	1942	88.93	98.3	24.2	0.0	100

3.2 Visualization

```r
# Reorder factor so Public School appears first for visual clarity
immun_1819$Type <- factor(immun_1819$Type,
                          levels = c("Public School", "Private School"))

boxplot(measles_pct ~ Type,
        data = immun_1819,
        main = "Measles Vaccination Coverage by School Type",
        xlab = "School Type",
        ylab = "Coverage (%)",
        col = c("lightblue", "mistyrose"))

# Add reference line for herd immunity threshold
abline(h = 95, lty = 2, col = "red", lwd = 1.5)
legend("bottomleft", legend = "95% herd immunity threshold",
       lty = 2, col = "red", lwd = 1.5, bty = "n", cex = 0.85)
```

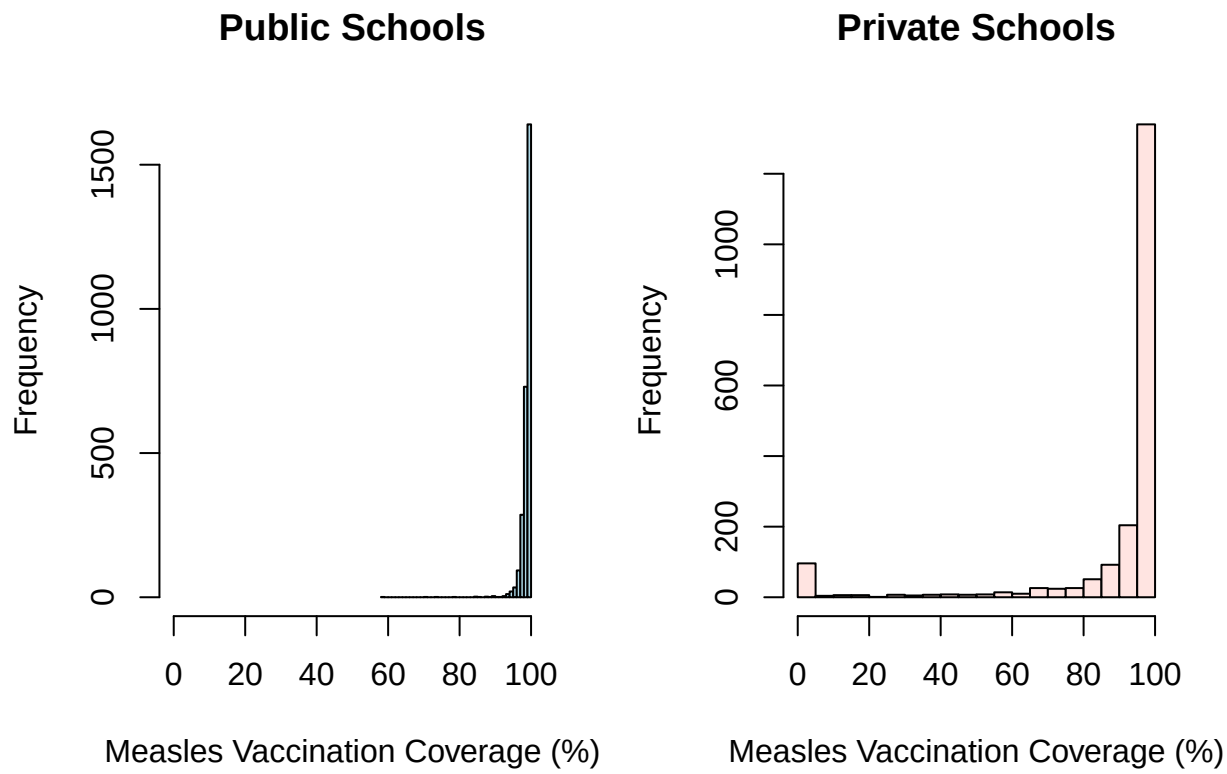


The boxplot indicates that public schools generally had higher and more consistent measles vaccination coverage than private schools. Both groups had median coverage near or above 95%, but private schools showed considerably greater variability and many more low-coverage outliers. The dashed reference line at 95% marks the approximate threshold for herd immunity; notably, a larger proportion of private schools fell below this threshold.

```
par(mfrow = c(1, 2))

hist(public_schools$measles_pct,
     main = "Public Schools",
     xlab = "Measles Vaccination Coverage (%)",
     col = "lightblue",
     breaks = 30,
     xlim = c(0, 100))

hist(private_schools$measles_pct,
     main = "Private Schools",
     xlab = "Measles Vaccination Coverage (%)",
     col = "mistyrose",
     breaks = 30,
     xlim = c(0, 100))
```




```r
par(mfrow = c(1, 1))
```

The histograms confirm that public schools had consistently high measles vaccination coverage, with most schools clustered near 100%. Private schools also had many schools near high coverage levels, but their distribution was considerably more spread out and included substantially more schools with low coverage. This pattern is consistent with greater variability and lower overall coverage in private schools.

3.3 Statistical Analyses

A Welch two-sample t-test showed that mean measles vaccination coverage differed significantly between public and private schools. Mean coverage was 98.85% in public schools and 88.93% in private schools. The difference was statistically significant ($t = -18.04$, $df = 1954.2$, $p < 2.2e-16$). The estimated mean difference was -9.92 percentage points (private minus public), with a 95% confidence interval from -11.00 to -8.84.

The Wilcoxon rank-sum test yielded a consistent conclusion ($W = 2{,}146{,}194$, $p < 2.2e-16$), confirming that the difference in measles vaccination coverage between public and private schools was statistically significant even under the non-parametric approach that does not assume normality.

Taken together, these results indicate that private schools had substantially lower measles vaccination coverage than public schools during the 2018–2019 school year.

4 Conclusions

The results support the research question by demonstrating that measles vaccination coverage differed significantly between public and private schools in New York State during the 2018–2019 school year. Private

schools had substantially lower mean coverage (88.93%) than public schools (98.85%), consistent with prior literature showing higher exemption rates in private school settings.

From a public health perspective, this difference is meaningful because measles is highly contagious and outbreaks can occur when vaccination coverage falls below the threshold required for herd immunity. The lower and more variable coverage observed in private schools suggests the presence of pockets of susceptibility where vaccine-preventable outbreaks are more likely to occur.

Several limitations should be noted. First, this analysis used school-level aggregate data; individual-level vaccination status was not available, and findings cannot be attributed to specific students or subgroups. Second, the analysis did not control for potential confounders such as school size, geographic region, or socioeconomic characteristics, which may partly explain the observed differences. Third, the data excluded New York City public schools, which limits generalizability to the full state population.

Overall, these findings suggest that school type is an important factor to consider in vaccination outreach and prevention efforts. Targeted public health interventions directed at private schools may be warranted to close the coverage gap and reduce the risk of measles outbreaks.

5 AI Use Disclosure Statement

This document was completed with assistance from Claude. Claude was used to help recall R functions and to provide explanations and examples of their usage. All code and interpretations were reviewed and finalized by the author.

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Shaw, Jana, Boldtsetseg Tserenpuntsag, Louise-Anne McNutt, and Neal Halsey. 2014. "United States Private Schools Have Higher Rates of Exemptions to School Immunization Requirements Than Public Schools." *The Journal of Pediatrics* 165 (1): 129–33. https://doi.org/10.1016/j.jpeds.2014.03.039.